BudgetAdvisor (Group G12) - Optimization & DAA Report

0/1 Knapsack Problem: FIFO Branch & Bound, LC Branch & Bound, and DP Verification

SECTION 1

Project and Optimization Details

APPLICATION & GROUP **BudgetAdvisor (Group G12)**	SELECTED MODE **Investment Advisor**	DATASET NAME **Investment Portfolio Dataset**
AVAILABLE BUDGET **Rs. 10,000.00**	DATASET SOURCE & SIZE **Custom (8 of 24 max)**	RUN ID & TIMESTAMP **run-audit- | 2026-11-06 02:58:41 UTC**

Disclaimer: Investment mode uses an educational additive score model on fictional assets. Scores are not financial returns or investment advice.

SECTION 2

Input Dataset (8 Items - Investment Portfolio Dataset)

#	Item / Asset Name	Cost (INR)	Value / Score	Score/Cost	Status
1	Atlas index fund	Rs. 4,000.00	12.50	0.0031	Excluded
2	Cedar green energy	Rs. 2,500.00	16.00	0.0064	Selected
3	Meridian technology	Rs. 3,500.00	18.00	0.0051	Selected
4	Harbor bond fund	Rs. 2,000.00	7.50	0.0037	Excluded
5	Orion healthcare	Rs. 3,000.00	14.00	0.0047	Excluded
6	Maple consumer fund	Rs. 1,500.00	9.00	0.0060	Selected
7	Summit infrastructure	Rs. 4,500.00	17.00	0.0038	Excluded
8	Willow small-cap fund	Rs. 2,200.00	15.50	0.0070	Selected

SECTION 3

Optimal Recommendation

RECOMMENDED STRATEGY **FIFO Branch & Bound**	MAXIMUM ACHIEVABLE SCORE **58.50 pts**	TOTAL EXPENDITURE **Rs. 9,700.00**
REMAINING BUDGET **Rs. 300.00**	SELECTED ITEMS COUNT **4 items**	OPTIMALITY STATUS **Proven Optimal**

#	Recommended Item / Asset	Individual Cost	Individual Value
1	Cedar green energy	Rs. 2,500.00	16.00 pts
2	Meridian technology	Rs. 3,500.00	18.00 pts
3	Maple consumer fund	Rs. 1,500.00	9.00 pts
4	Willow small-cap fund	Rs. 2,200.00	15.50 pts

SECTION 4

Algorithm Performance (FIFO vs. LC Branch and Bound)

Algorithm	Exec Time	Generated	Explored	Pruned	Frontier	Max Queue	Opt Val	Status
FIFO Branch & Bound	0.386 ms	39	39	19	0	17	58.50	Proven Optimal
LC Branch & Bound	0.212 ms	17	17	8	0	9	58.50	Proven Optimal

Node-Limit & Pruning Note: Configured search limit caps explored (visited) nodes at 50,000 (1,500 ms timeout). Pruned nodes represent the sum of upper-bound pruned nodes (node.bound < best_score) and infeasible budget-pruned nodes (node.cost > budget). Because each expanded node generates 2 children, generated nodes (1 + 2*expanded = explored + frontier) can exceed explored nodes when a search is capped.

Multi-Size Scaling Benchmark Summary (Source: Custom [Investment Portfolio Dataset], Base 8 items, Expanded to 24 items, Variations: Yes, Budget Rs. 10,000, Seed 42):

Size	Algo	Median Time	Generated	Explored	Pruned	Max Queue	Opt Val
5 items	FIFO	0.158 ms	19	19	9	9	48.00
5 items	LC	0.100 ms	11	11	5	6	48.00
10 items	FIFO	0.349 ms	63	63	31	19	61.00
10 items	LC	0.269 ms	39	39	19	20	61.00
15 items	FIFO	0.517 ms	131	131	65	20	66.50
15 items	LC	0.487 ms	101	101	50	51	66.50
20 items	FIFO	0.550 ms	141	141	70	24	69.00
20 items	LC	0.530 ms	111	111	55	56	69.00
24 items	FIFO	0.640 ms	159	159	79	45	72.50
24 items	LC	0.359 ms	49	49	24	25	72.50

Note on Custom Datasets: Datasets exceeding the original custom item count are expanded using reproducible variations derived from the original dataset. These additional entries are generated for algorithm scalability testing and are not independently entered real-world products.

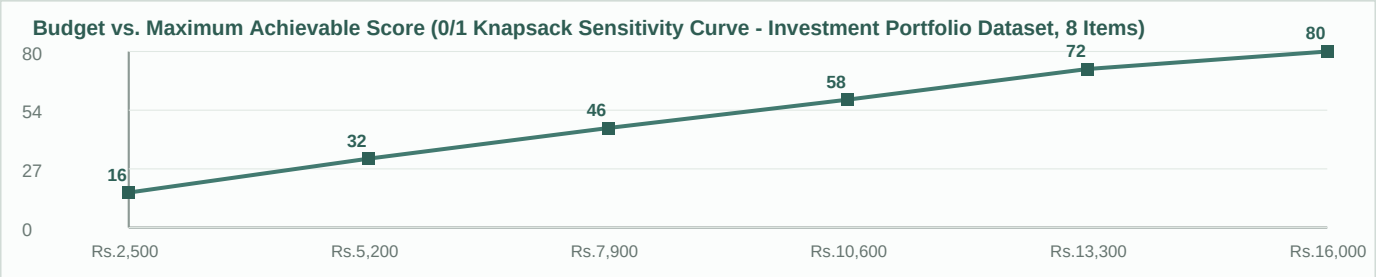
SECTION 5

Independent Dynamic Programming Verification

DP BASELINE SCORE **58.50 pts (Rs. 9,700.00)**	DP STATES EXPLORED **86 (0.062 ms)**	DP COMPLETION STATUS **Proven Optimal**
FIFO CORRECTNESS STATUS **Verified Match**	LC CORRECTNESS STATUS **Verified Match**	TRI-METHOD CONSENSUS **All 3 Return Identical Value**

SECTION 6

Budget Sensitivity / What-If Analysis



Tested Budget	Max Score	Expenditure	Remaining	Items	LC Time	Verification
Rs. 2,500.00	16.00 pts	Rs. 2,500.00	Rs. 0.00	1 items	0.25 ms	Verified
Rs. 5,200.00	31.50 pts	Rs. 4,700.00	Rs. 500.00	2 items	0.19 ms	Verified
Rs. 7,900.00	45.50 pts	Rs. 7,700.00	Rs. 200.00	3 items	0.24 ms	Verified
Rs. 10,600.00	58.50 pts	Rs. 9,700.00	Rs. 900.00	4 items	0.24 ms	Verified
Rs. 13,300.00	72.50 pts	Rs. 12,700.00	Rs. 600.00	5 items	0.15 ms	Verified
Rs. 16,000.00	80.50 pts	Rs. 15,700.00	Rs. 300.00	5 items	0.26 ms	Verified

- Budget expansion increased achievable score: Increasing the available budget from Rs. 2,500.00 to Rs. 16,000.00 (+Rs. 13,500.00) improved the maximum achievable score from 16.00 to 80.50 (+64.50 points), expanding the optimal selection from 1 to 5 items.

- Budget threshold for high-value item "Meridian technology": Budgets up to Rs. 2,500.00 are insufficient to include "Meridian technology" (cost Rs. 3,500.00, score 18.00). A minimum budget of Rs. 3,500.00 is required before this item becomes individually feasible.

SECTION 7

Conclusion

All three methods (FIFO, LC, and sparse Dynamic Programming) independently confirmed the global optimum of 58.50 points at a total cost of Rs. 9,700.00 (leaving Rs. 300.00 unspent from the Rs. 10,000.00 budget across 4 verified items). LC Branch and Bound explored 17 nodes and pruned 8 nodes in total, including 4 bound-pruned nodes and 4 infeasible nodes, compared to 39 explored nodes for FIFO Branch and Bound.